

Very Very Very

Important for
Interview and
Concept jobs

June 15

Python

Comprehension

Concept

Jobs

The process of creating new sequence object by using existing sequence object is called comprehension

It is basically a way of writing a entire code in a single line to generate a sequence object which can be list, dictionary, set, generator

In Python we have 4 types of "comprehension" They are,

- 1) List comprehension
- 2) Set comprehension
- 3) Dictionary comprehension
- 4) Generator comprehension

① List comprehension

→ List comprehension provide a concise block to create a new ^{list} object

→ It consists of square brackets containing an expression followed by for clause then zero or more

Syntax -

[expression ,
mandatory

optional for or

Ex Example

$[x * x \text{ for } x \text{ in range}(10)]$ - list

$\{x * x \text{ for } x \text{ in range}(10)\}$ - set

$\{x : x * x \text{ for } x \text{ in range}(10)\}$ - dictionary

June 16 all Important Programs

Important Programs

1. Generate a list of square of numbers from 1 to 10

Squares = $[x * x^2 \text{ for } x \text{ in range}(1, 11)]$

print (squares)

output :- $[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]$

2

Get all even numbers between 1 and 20

evens = $[x \text{ for } x \text{ in range}(1, 21) \text{ if } x \% 2 == 0]$

$x \% 2 == 0$

print (evens)

output $[2, 4, 6, 8, 10, 12, 14, 16, 18, 20]$

③ Get all odd numbers between 1 and 20

```
odds = [x*x for x in range(1, 21)]
```

```
print(odds)
```

output:- 1, 3, 5, 7, 9, 11, 13, 15, 17, 19

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④ Convert a list of strings to uppercase

```
words = ["apple", "banana", "cherry"]
```

```
upper-words = [w.upper() for w in words]
```

```
print(upper-words)
```

output:- ['APPLE', 'BANANA', 'CHERRY']

⑤ Extract all vowels from a string

```
text = "comprehensions"
```

```
vowels = [ch for ch in text if ch in "aeiou"]
```

```
print(vowels)
```

output: ['o', 'e', 'e', 'i', 'i']

⑥ Create a list of tuples (number, square)
for numbers from 1 to 5

Pairs = $[x, x \times x^2]$ for x in range(1,6)

print (pairs)

output $[(1,1), (2,4), (3,9), (4,16), (5,25)]$

⑥ Get unique vowels from a sentence

sentence = "python comprehensions are powerful"

unique-vowels = $\{ch \text{ for } ch \text{ in text if } ch \text{ in 'aeiou'}\}$

print (unique-vowels)

output $['e', 'a', 'i', 'o', 'u']$

⑦ Find all unique words from a list, ignoring case

words = $['Apple', 'pine Apple', 'Banana', 'BANANA']$

unique-words = $\{w.lower() \text{ for } w \text{ in words}\}$

print (unique-words)

output $\{'banana', 'apple'\}$